

In-air Charging of Unmanned Aerial Vehicles

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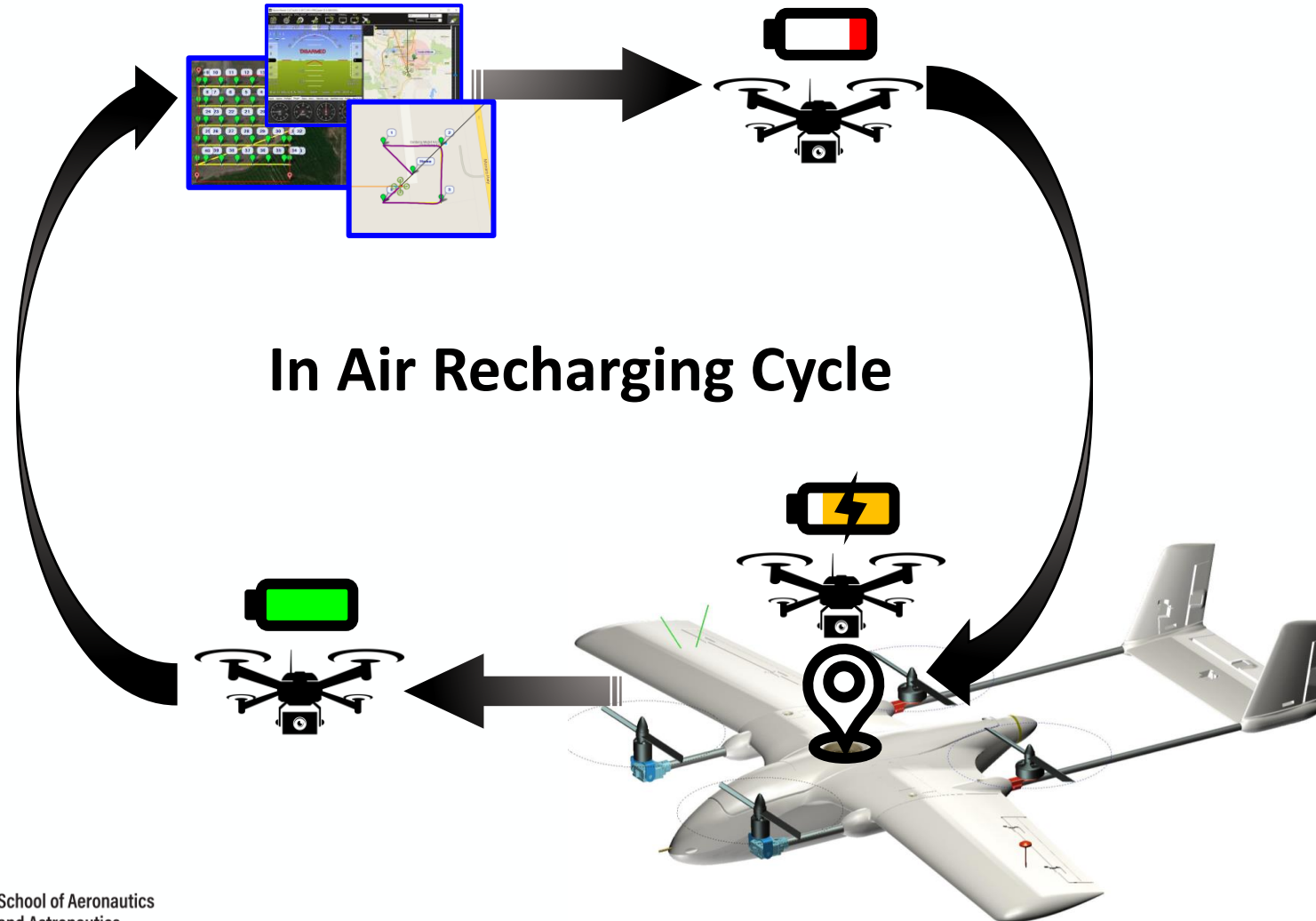
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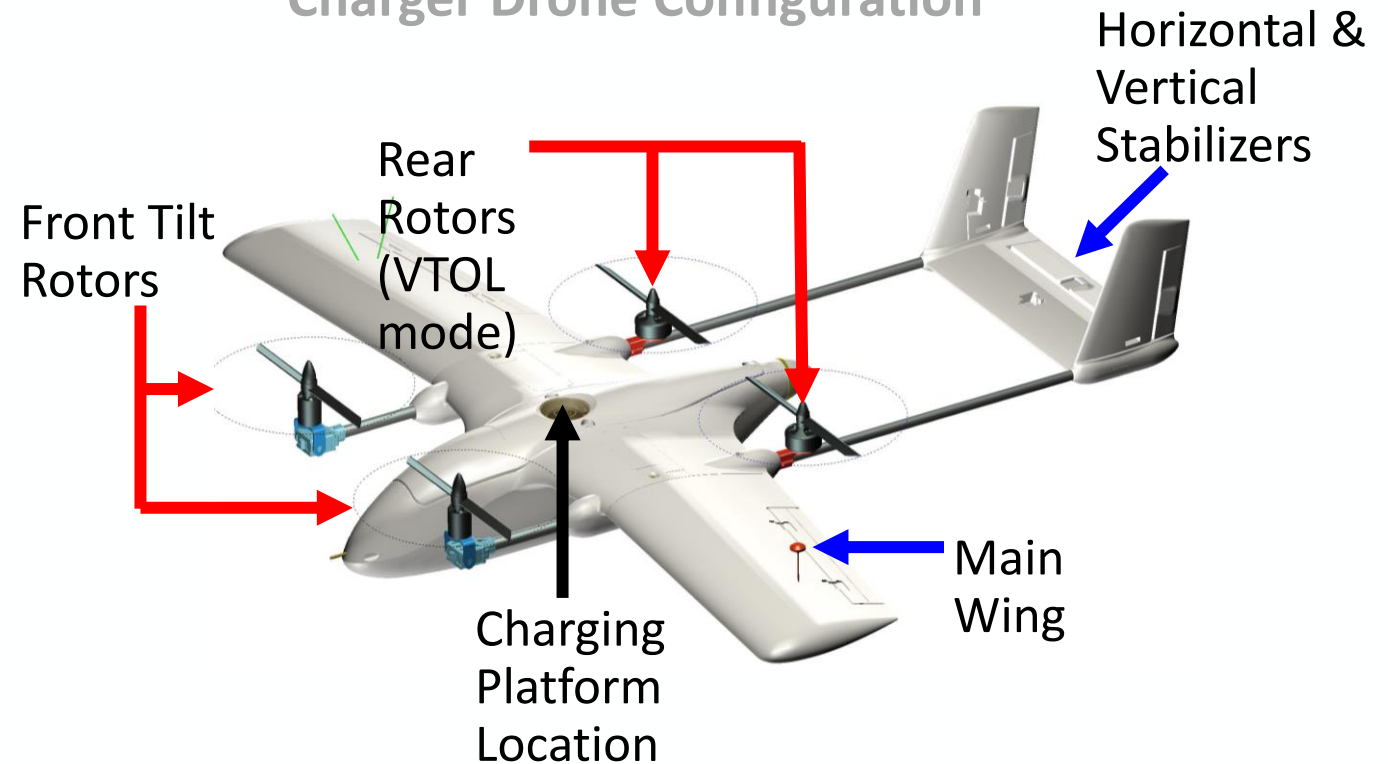
Introduction/Background



What we are researching

- Develop and integrate an aerial recharging platform for a specific drone.
 - Entails cohesive landing and charging designs.
- Develop a tracking platform to connect the charging drone and the drone to be charged.
- Ensure uninterrupted, autonomous flight during the drones cycle.

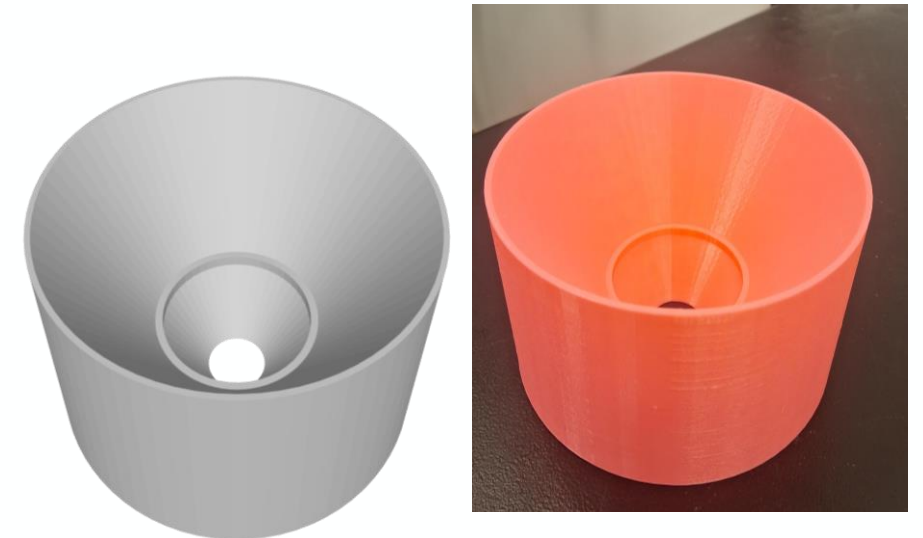
Charger Drone Configuration



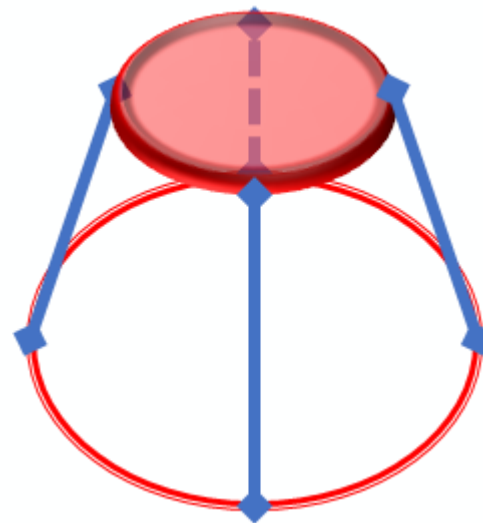
Methods

In-Air Charging Design

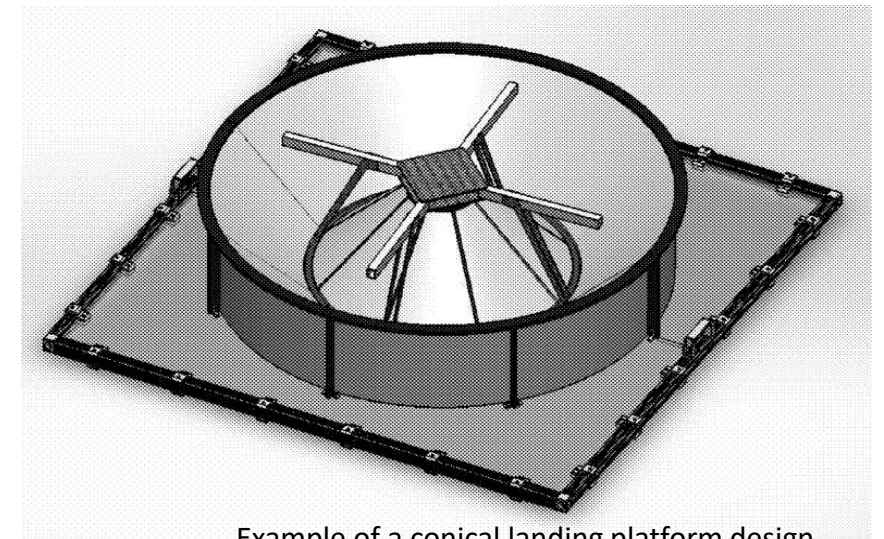
- Goal:
 - Design & prototype quick charging platform
- Key features:
 - Landing legs equipped with funnels guide charging cables to ports
 - Minimizes the need for precise landing maneuvers
- Advantages:
 - Enhanced flexibility in drone landing operations
 - Reduces dependency on precise landing accuracy
 - Ensures consistent and reliable charging for drones



Landing funnel CAD design & the 3D printed funnel



Landing gear that will fit to the funnel

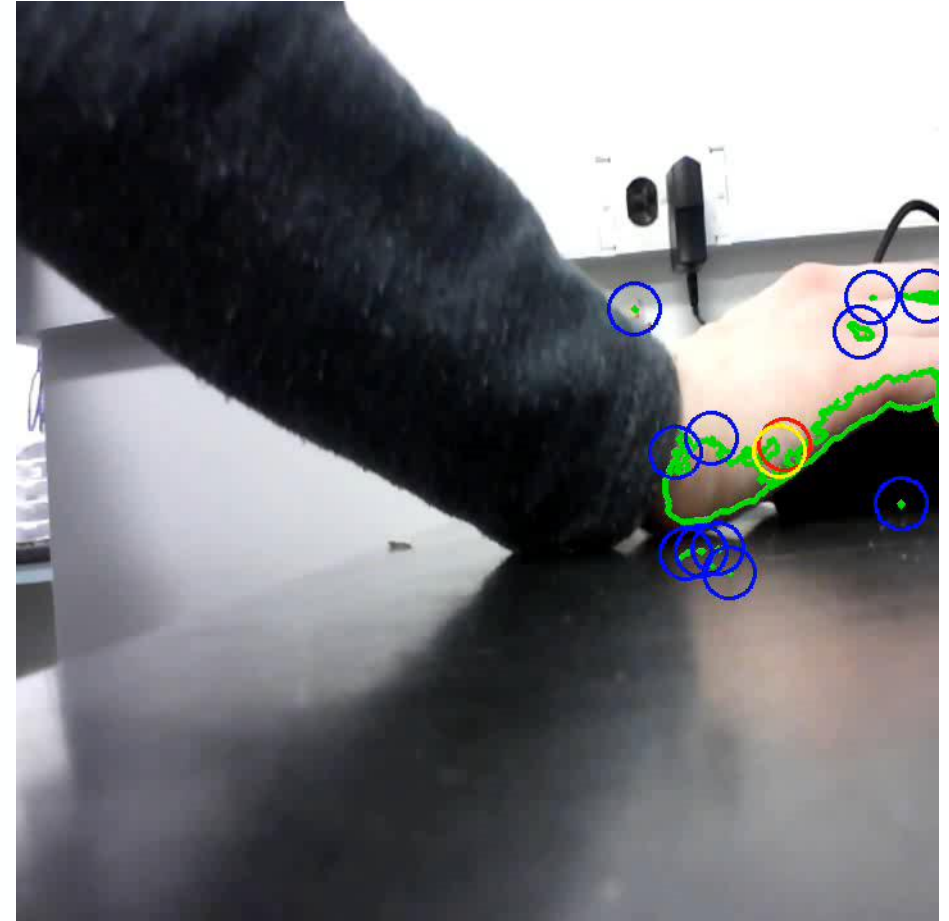


Example of a conical landing platform design

Methods

Autonomous Operations – Vision Tracking

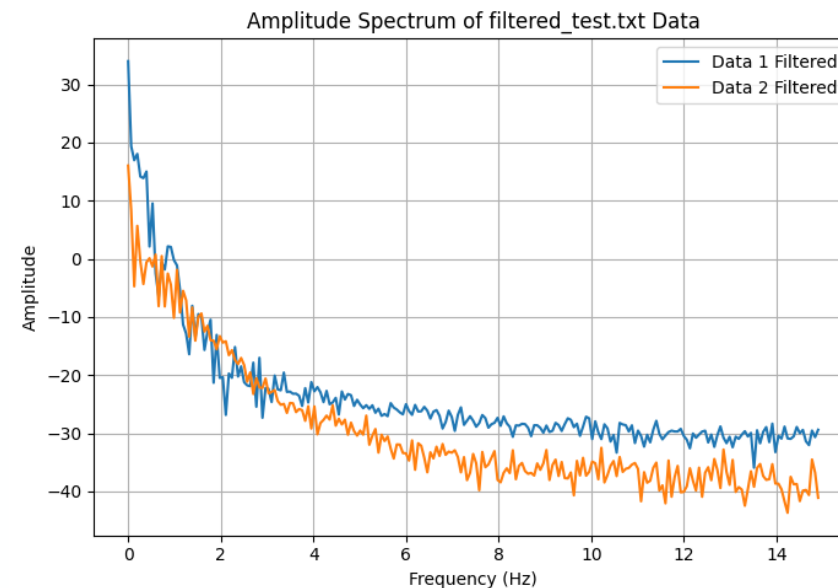
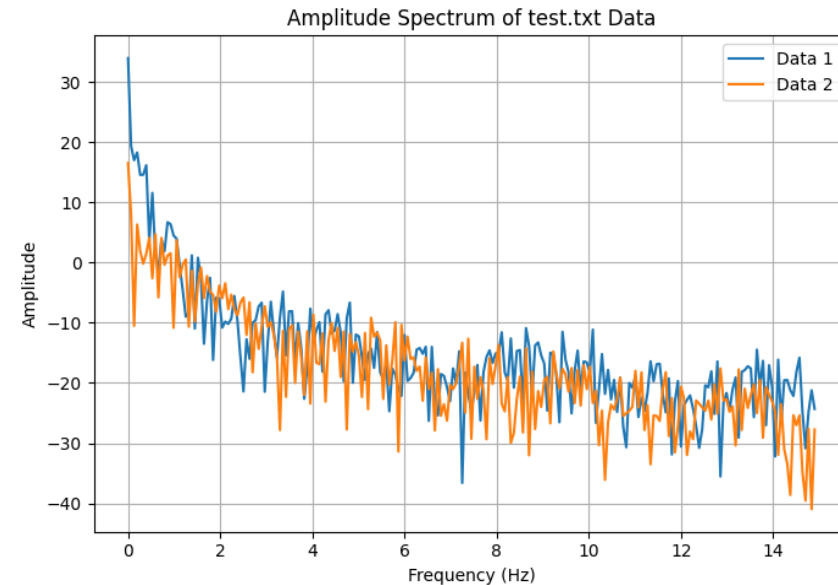
- Vision-based target detection algorithm
 - Target detection measured center of drone and distance to drone.
 - Steps:
 - Mask
 - Compare centroid to average
 - Calculate distance
 - Calculate mean of value group
- These values are filtered and sent to the main drone via Mavlink.



Methods

Autonomous Operations – Data Filtering

- X & Y centroid values are collected with lots of noise.
 - Designed and implemented a Butterworth filter to reduce the effect of noise at useful frequencies.
- Compare measured location to previous locations to eliminate outlying data.
 - Tested for multiple distances.



Results

3/26

- Individual successful testing of large drone, small drone, and charging platform.
 - Successful 6DOF Modeling of flight dynamics simulation ability.
 - Centroid estimation and filtering algorithm have been successfully integrated
 - Integrating charging and landing platform.

Discussion/Wrap-Up

Next Steps:

- Integration of each individual subsystem
- Looking to charge 2 cycles ~10-minute test
- Iterate, Iterate, Iterate

Future Direction

Gateway to Aerospace development

- Extended flight durations
- Continuous mission capability
- Improved efficiency
- Technological advancement
- Environmental benefits



Thank You



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